

Armond Welch

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EXPERIENCE

MedicalBlox, LLC (Startup/Initiative)

Tulsa, OK

Developer — Remote

June 2021-June 2025

- Co-developed electronic medical record (EMR) software focused on reducing siloing and enhancing interoperability across healthcare systems reporting directly to a radiation oncologist (founder).
- Contributed to full-stack development as a software developer, building responsive front-end patient portal with HTML, CSS, and JavaScript and implementing server-side logic in Node.js.
- Used Python and Go to integrate a permissioned blockchain network with Hyperledger Fabric and a NoSQL database to transfer JSON-formatted medical record data to and from an EMR sandbox via REST APIs.
- Implemented certificate-based identity management, TLS-encrypted communication, and channel-based access control to ensure secure transfer of medical records.

Level42AI

Troy, NY

RPI Capstone Project — Remote

October 2023-May 2024

- Worked with startup company Level42AI to enhance proprietary Vibrome wearable device, a multimodal vibroacoustic and capacitive biosensing platform for non-invasive cardiac monitoring.
- Led 4-person cohesive team in developing and implementing a dynamic noise filtering algorithm to improve signal fidelity across multiple modalities with Python, C++, and PSoC creator (embedded).
- Supported integration of capacitive EKG electrodes (gel-free, air-gap operation) with vibroacoustic sensors for simultaneous acquisition of EKG, BCG (Ballistocardiogram), and pressure-volume loop data. Also drafted designs for EEG analysis using capacitive electrodes but was not implemented.
- Contributed to signal processing optimization for real-time cardiac waveform analysis.

VDO Cardiology

Briarcliff Manor, NY

Medical Scribe — Remote

June 2020-December 2023

- Provided real-time, high-quality remote clinical documentation for an electrophysiologist, ensuring accurate and efficient medical record keeping.
- Took on practice management role to coordinate patient testing, office correspondence, and appointments during COVID pandemic.
- Gained familiarity with industry standards of care and detailed clinical note structuring under real-time physician guidance.

EDUCATION & ACTIVITIES

Rensselaer Polytechnic Institute

Troy, NY

Bachelor of Science in Biomedical Engineering

Class of 2024

- Undergraduate Research Program: Classification of necrotic tissue using convolutional neural networks; Presented findings at the AAPM 66th Annual Meeting & Exhibition (July 2024)
- Alto Saxophonist, RPI Wind Symphony (2023–2024)

SKILLS: Java, Go, Javascript, Node.js, Kubernetes, Docker, Containerization, Databases, Network security, authentication, REST APIs, servers, async data transfer, problem solving, communication.